

Detector-coupled Monte Carlo estimate of background and unresolved-line sensitivity for a balloon-borne 511 keV Laue-lens TES telescope

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Abstract Background determines the sensitivity of a balloon-borne 511 keV Laue-lens telescope with a transition-edge-sensor (TES) microcalorimeter. We use an end-to-end Geant4/MEGAlib chain with three physically distinct inputs: focused celestial signal, a corrected-total-kinetic-energy atmospheric field containing all eight prompt particle families, and production-position-sampled activation decays. The broadband gamma field already contains the atmospheric annihilation feature, so it is counted once rather than supplemented by a monoenergetic component. Every stream receives the same 510.58–511.42 keV window, active anticoincidence, and topology/field-of-view selection. In the SG3 reference geometry the day-15 mature background is $5.1280 \times 10^{-2} \text{ s}^{-1}$, and 32.18% of its selected delayed rate originates in the dilution-refrigerator/mixing-chamber and staged cold plates. The source map motivates the SH3 geometry, which moves the six-layer TES focal plane into a lateral aluminium chimney surrounded by active BGO. The selected cold-stage share then falls to 3.92%. The selected effective area changes only from 15.12324 ± 0.04492 to $15.08544 \pm 0.04503 \text{ cm}^2$, whereas the day-15 mature background falls from 5.1280×10^{-2} to $9.280 \times 10^{-3} \text{ s}^{-1}$. At a top-of-atmosphere line flux of $10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$, the SH3 day-15 signal rate is $9.7904 \times 10^{-4} \text{ s}^{-1}$. A fixed-45°-elevation 20 d trajectory fold gives 1678.14 signal and 15551.68 background counts, Gaussian and Asimov significances of 13.46 and 13.22, and a Gaussian 3σ threshold of $(2.229 \pm 0.209) \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$. Relative to SG3, SH3 retains 99.75% of the effective area, reduces the background to 18.10%, and improves the minimum resolvable flux by a factor of about 2.43.

Keywords Gamma-ray instrumentation, 511 keV annihilation line, transition-edge sensors, Laue lens, balloon-borne telescope, activation background

1 Introduction

Galactic 511 keV annihilation radiation is a key observational probe of the production, propagation, and annihilation environments of Galactic positrons. Its spectrum contains a two-photon line near 511 keV and an ortho-positronium continuum at lower energies. Positrons annihilate directly with electrons or form para-positronium, producing two photons near 511 keV, whereas ortho-positronium decays into a three-photon continuum. The line intensity, profile, and positronium continuum therefore constrain the medium in which positrons slow down and annihilate [1, 2]. Under an approximate steady-state assumption, the total Galactic 511 keV flux measured by INTEGRAL/SPI corresponds to an annihilation rate of a few $\times 10^{43} \text{ e}^+ \text{ s}^{-1}$ [1, 3]. Proposed positron sources include β^+ -unstable nuclei synthesized in stellar explosions, compact objects capable of producing electron-positron pairs, and more speculative particle-physics scenarios in which annihilating or decaying light dark matter injects low-energy positrons. Existing observations do not uniquely determine the dominant contribution [1, 4, 5].

Wide-field observations have established the diffuse morphology of the line. INTEGRAL/SPI finds a bright central bulge and a weaker disk, and a recent 20-year analysis recovers central, broad-bulge, and disk components while the evidence for additional correlated structures remains marginal [4, 6, 7]. The 2016 COSI balloon flight independently detected the bulge and found evidence that the emission is more extended than a single point source [3, 8]. Spectroscopy also constrains the annihilation medium and kinematics: the SPI spectrum can be

decomposed into a narrow component with about 1.3 keV FWHM and a broad component with about 5.4 keV FWHM, and spatial variations in centroid energy and width have been sought across the inner Galaxy [2, 9]. The annihilation map, however, records where positrons slow down and annihilate, not necessarily where they were created. Propagation through a magnetized, multiphase interstellar medium means that diffuse morphology and spectroscopy alone cannot recover the dominant source class or the complete pre-annihilation transport history [1].

Wide-field Compton and coded-mask instruments are well suited to the total bulge and disk emission. A complementary pointed instrument is needed to test whether the central region also contains an unresolved central enhancement, compact sources, or measurable line-kinematic structure. A detection of a point-like or narrowly concentrated feature would show that part of the annihilation is spatially localized; a non-detection would constrain a compact component without excluding source populations whose positrons escape and annihilate diffusely. An INTEGRAL/IBIS compact-source search found no Galactic 511 keV point sources and set a 2σ upper limit of about $1.6 \times 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$ for a ~ 10 Ms Galactic-centre exposure [10]. This limit sets both the comparison scale of earlier compact-source searches and a sensitivity target for a new narrow-field pointing instrument.

Laue optics offer one route to such pointed observations by using transmission-mode Bragg diffraction to focus photons from a known direction onto a small focal plane [11, 12]. Concentrating the flux collected over a large aperture onto a smaller sensitive detector area can improve point-source sensitivity when detector-related background dominates [13]. The focal plane must then combine a narrow energy response with sufficient stopping efficiency at 511 keV. Transition-edge-sensor (TES) microcalorimeters use the steep resistive transition of a superconductor to measure small temperature changes from deposited energy and can provide sub-keV resolution at 511 keV [14, 15]. An array covers the focal spot, while thick absorbers and a multilayer layout raise the full-energy efficiency. Laue focusing compresses the accepted phase space, and the TES response permits a narrow line window and detailed line-profile measurements. The resulting instrument trades field of view for a concentrated beam and high-resolution spectroscopy, complementing rather than replacing wide-field bulge and disk measurements.

However, the intrinsic high energy resolution of the TES detector and the focusing performance of the optics are not by themselves sufficient to establish the point-source resolving capability of the complete detection system [14, 16]. Both the prompt background spectrum generated by cosmic-ray energy deposits in the detector's sensitive volume and the secondary delayed background spectrum produced by cosmic-ray activation of the detector and surrounding materials affect the system's ability to resolve a scientific source [17, 18].

This work aims to quantify the high-altitude detection performance of a balloon-borne Laue-lens TES-array telescope system for a 511 keV point source at the Galactic center, while characterizing the origin and distribution of the background to guide further design optimization.

1.1 Science and instrument requirements

The simulated instrument is designed to detect either 511 keV point-source emission or an unresolved central-excess component in the Galactic-center region, neither of which has been detected by previous instruments. INTEGRAL/SPI measurements show that the 511 keV sky is dominated by diffuse bulge and disk emission [4, 7]. De Cesare's dedicated search for compact sources with INTEGRAL/IBIS found no Galactic 511 keV point sources and reported a 2σ upper limit for a 10 Ms exposure toward the Galactic center [10]. To compare this limit with the 3σ sensitivity reported here, we linearly rescale the published limit under the Gaussian approximation, obtaining an IBIS-based 3σ -equivalent top-of-atmosphere upper limit of $2.4 \times 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$. At the altitude on flight day 15, the remaining vertical atmospheric column depth is 3.461 g cm^{-2} . For the instrument axis, designed for a fixed elevation of 45° , the plane-parallel approximation gives a slant column depth of 4.895 g cm^{-2} ; using an effective dry-air mass attenuation coefficient of $0.08736 \text{ cm}^2 \text{ g}^{-1}$, consistent with NIST XCOM data near 511 keV, gives a transmission of $T_{15,45} = 0.6520$. Applying this transmission defines the single observational reference flux at the payload plane, $F_{0,\text{ref}} = (2.4 \times 10^{-4}) \times 0.6520 = 1.565 \times 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$. This reference flux provides a benchmark for the minimum resolvable flux achievable by the simulated telescope system within an acceptable flight duration, as reported below. Compared with the wide-field spectroscopic and coded-mask searches conducted with INTEGRAL, the payload modeled here introduces Laue focusing to improve the

point-source response and uses a high-energy-resolution, multilayer thick-absorber TES microcalorimeter array as the focal-plane detector to improve its capability for 511 keV line measurements [14, 19].

This work first considers a balloon payload as an early test platform, with the expectation that the approach can later be extended to a satellite observing platform.

This objective imposes four requirements on the instrument and the simulation. First, the telescope must focus the 511 keV photons collected by a large-aperture lens onto a compact focal plane, thereby increasing the ratio of the collection area of the focusing optics to the sensitive area of the detector; this is a narrow-field pointed configuration, not a means of increasing sky coverage. Second, the focal-plane detector must provide a sub-keV energy response sufficient to resolve fine line structure. Third, the detector and shielding model must suppress atmospheric, activation, and side-entry backgrounds. Fourth, the simulation chain must treat prompt background, delayed activation, focused signal, vetoes, event topology, and mission-time variation consistently, so that signal and background are evaluated using the same detector-selection criteria.

The remainder of this paper is organized as follows. Section 2 presents the detector and cryostat mass model; Section 3 describes the simulation framework, including the simulation workflow, optics coupling, background sources, event selection, and mission-time trajectory folding (Section 3.5); Section 4 presents the post-selection rates and sensitivity results together with the normalization, convergence, and boundary checks; and Section 5 contains the discussion and conclusions.

2 Detector and cryostat mass model

This study uses the current SG3 detector–cryostat geometry as the auditable reference. It contains a balloon-borne six-layer TES array below the mixing chamber and staged cold plates of a compact dilution refrigerator (DR), together with thermal and magnetic shields, copper supports, side-entry windows, a tungsten collimator, active scintillators, and cryogenic service structures [14, 20]. Figure 1 shows the instrument-frame section used to connect the reference focal-plane placement to the selected-event background analysis. The SH3 side-chimney geometry introduced in Section 4.2 retains the same detector, optical, response, and source contracts so that the geometry comparison is like-for-like.

The TES detector concept uses an absorber-coupled AlMn-film structure developed in our laboratory. The transition temperature of AlMn can be tuned through its composition and annealing process; the material is suitable for array fabrication and has relatively low magnetic-field sensitivity [21, 22]. In the present conceptual design, the TES thermometer is an approximately 200 nm-thick AlMn film with a nominal Mn atomic concentration of 2000 ppm, whereas the γ -ray absorber is a millimetre-scale metal body. The absorber consists of $1.5 \times 1.5 \times 3.0 \text{ mm}^3$ Ta pixels; the high atomic number of Ta increases the interaction efficiency for 511 keV photons. At 511 keV, a single 3.0 mm Ta layer has an estimated interaction efficiency of about 49%, and the corresponding six-layer efficiency is close to unity. In the current geometry, adjacent Ta array layers have a center-to-center spacing of 12 mm; this spacing was chosen with a subsequent veto based on Compton-kinematic track reconstruction in mind. Ta was also selected for its low volumetric heat capacity as a superconductor at cryogenic temperatures (an order-of-magnitude estimate of $10^{-2} \text{ J K}^{-1} \text{ m}^{-3}$ is used here) and its relatively high photoelectric-to-Compton interaction ratio. From the NIST XCOM cross sections near 511 keV, this ratio is approximately 0.731 [23]. In the detector-response model, we use our laboratory experience with a previously developed X-ray TES [21] to adopt $\Delta E_{\text{FWHM}} \simeq 420 \text{ eV}$ at 511 keV as the estimated energy resolution and detector-response post-processing input.

Because the TES film and Ta absorber differ greatly in dimensions and mass, and because the TES film itself contributes negligibly to interactions at 511 keV, the transport geometry uses the Ta absorber as the principal sensitive volume of each TES pixel. The explicitly modelled sensitive volume comprises six layers, each containing 376 Ta absorber pixels, for a total of 2256 pixels. Each pixel is 1.5 mm long and wide and 3 mm thick. A Si substrate proxy is placed below each Ta pixel; because detailed thermal conduction is not modelled, Si represents the SiO_2 – SiN_x –Si substrate. The TES film, wiring, and readout details are not constructed layer by layer as explicit interaction volumes, but enter the model through the assumed energy response and service-mass proxies. Defined independently of the response FWHM, the retained science-window analysis selection uses the 511 keV line window $W_{511} = 511 \text{ keV} \pm 420 \text{ eV}$, or 510.58–511.42 keV.

The SG3 TES array is mounted on copper supports immediately below the staged cold plates. Along the optical path are successive aluminium thermal-shield windows, a beryllium vacuum window, and a tungsten

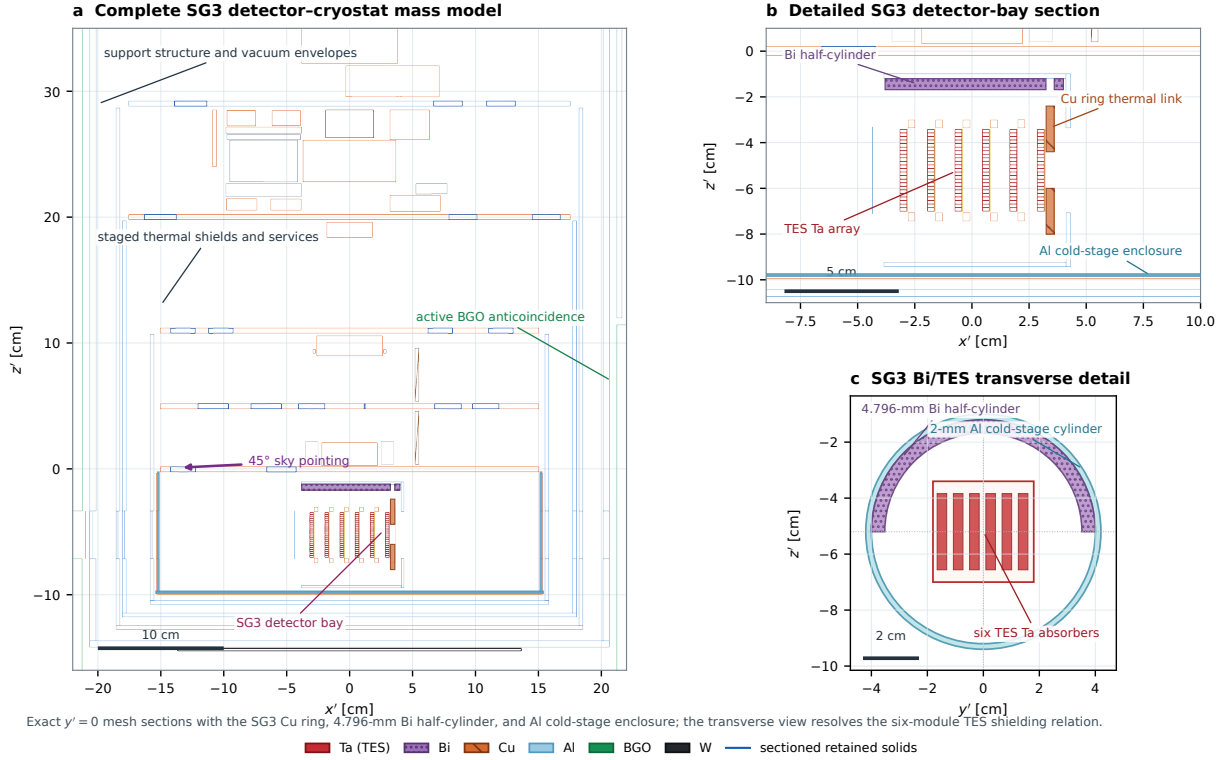


Figure 1: Current SG3 reference detector-cryostat mass model. (a) Exact global $y' = 0$ mesh section containing the support structure, vacuum envelopes, nested thermal shields, active BGO, cold plates, and detector bay; the arrow marks the fixed-45° sky-pointing direction. (b) Enlarged detector-bay section with the six TES planes, Cu ring thermal link, Bi half-cylinder, and Al cold-stage enclosure. (c) Section normal to the cylinder axis, resolving the 4.796-mm Bi half-cylinder, 2-mm Al cylinder, and six TES Ta absorbers.

multihole collimator aligned with the TES pixels. Boolean apertures through the surrounding shells define the side-entry beam path, and the complete instrument frame is rotated by 45° so that the local aperture points 45° above the horizon.

The optimized SH3 layout preserves the six TES layers and the upstream optical acceptance but translates the focal plane laterally into a cylindrical aluminium chimney. The chimney has inner and outer radii of 10.75 and 11.05 cm, a 3 mm wall, and a common local axis at $z' = -2.80$ cm. A compact tungsten frame supports the focal plane, and active BGO covers the chimney side, the front optical annulus, and the rear cold-port annulus. Energy deposits in the three active BGO detector volumes are summed event by event. Their native trigger reference is 80 keV, while the common offline anticoincidence gate used below accepts only groups with less than 50 keV summed BGO energy.

The mass-model section in Figure 1 is drawn from the retained triangle mesh rather than from a reduced detector cartoon. The global panel therefore keeps the vacuum enclosure, support shell, nested thermal shields, service penetrations, active BGO, cold plates, and detector bay on one common instrument coordinate system. The near-field panel resolves the components that matter directly for the 511 keV background: the six Ta absorber planes, their copper support and heat-sink connection, the aluminium cold enclosure, and the bismuth member above the focal plane. In SG3 the latter is a 4.796-mm-thick upper half-cylinder nested immediately inside the retained 2-mm aluminium cylinder. This construction preserves a continuous optical opening while avoiding the unphysical interpretation of a flat, infinite absorber plate.

The three panels also separate two geometrical questions that enter the later analysis. The global section establishes which massive payload structures can produce or scatter particles into the cryostat, whereas the two near-field sections establish which activated structures subtend the TES after production. The selected-event provenance analysis uses the generated volume names and coordinates associated with this same geometry, not

hand-assigned radial regions. Consequently, a delayed event attributed to a mixing-chamber plate, a copper ring, or a TES-near member can be placed back into the geometry without changing coordinate systems. This source–geometry correspondence is the basis for distinguishing activity that is merely present in the payload from activity that is efficiently coupled to the science window.

SG3 and SH3 retain the same six sensitive planes, pixel dimensions, response model, entrance aperture, and optical-axis convention. The controlled change is the relationship between the focal plane and the cold structures. In SG3 the TES lies underneath their projected area. In SH3 it lies along the lateral chimney axis, behind a compact local frame and within an active-BGO envelope. Keeping the detector and optical definitions fixed is essential: otherwise a reduction in background could not be separated from a simultaneous loss of focused acceptance.

3 Simulation framework

Detector-coupled Monte Carlo transport through detailed mass models is a mature method for predicting the response and background of gamma-ray spectrometers [24]. The optics branch transports an on-axis 511 keV point source through the Laue lens and records the focal-plane phase space. The detector branch then treats three physical chains independently: the resulting focused EventList, the corrected broadband atmospheric field, and radioactive decays sampled from their recorded production positions. Activation production is the source-building stage of the delayed chain rather than a counted prompt component. All three chains receive the same pixel response, science window, active BGO anticoincidence, and topology/field-of-view selection. Their selected rates are placed on a common time axis and folded through the same 81-node trajectory.

Figure 2 retains the mature source-to-response logic while making the present source contract explicit. The corrected broadband gamma tables include the atmospheric annihilation feature, so no independent monoenergetic atmospheric line is added. The source decomposition and mission fold then connect the SG3 reference diagnosis to the SH3 side-chimney comparison. Day-scale flights sample a continuum of atmospheric and geomagnetic states, so the trajectory fold and its five common-time anchor calculations are developed in Section 3.5.

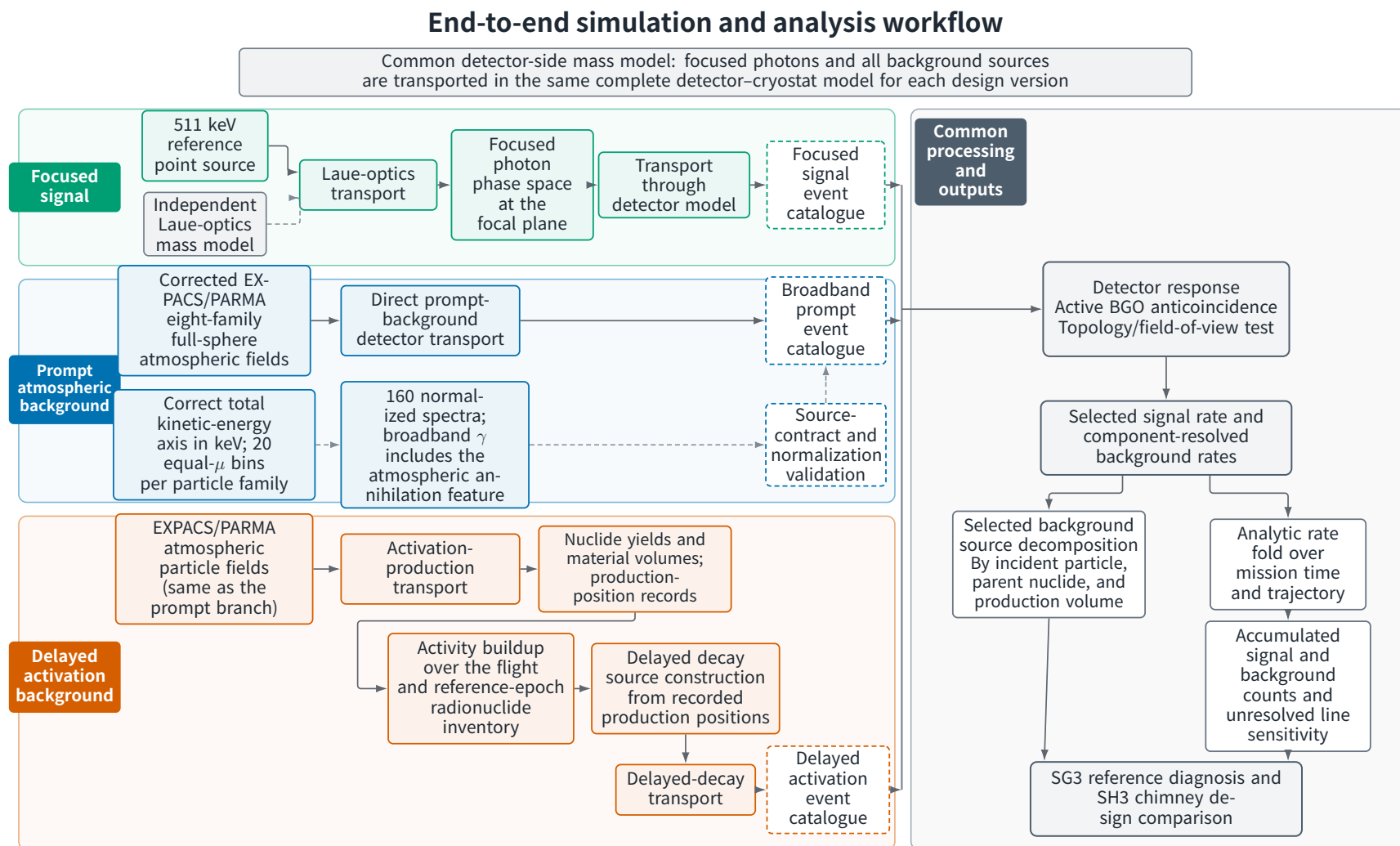


Figure 2: End-to-end simulation and analysis workflow. Focused signal, the corrected eight-family atmospheric field, and production-position-sampled activation decays are transported separately and then subjected to the common detector response, science window, active BGO anticoincidence, and topology/field-of-view selection. The selected rates enter source decomposition and the 81-node mission fold used for the SG3–SH3 comparison.

3.1 Laue optics and detector-coupled signal transport

This simulation adopts a laue-lens system based on mosaic Ge(111) crystals [11, 12]. Through optical design, it realizes a 10 m focal length, a 511 keV single-ring reference response, and an effective area of 20.1 cm² at 511 keV; the effective area is calculated as

$$A_{\text{eff}}(E) = A_{\text{inc}} \frac{N_{\text{fp}}(E)}{N_{\text{inc}}(E)} \quad (1)$$

where A_{inc} is the Monte Carlo incident sampling area, $N_{\text{inc}}(E)$ is the number of incident photons, and $N_{\text{fp}}(E)$ is the number of photons satisfying the focusing/focal-plane interception condition. The mosaic system is adopted to make the Monte Carlo calculation convenient while satisfying the expected focusing performance; other candidate schemes are also possible for actual engineering adaptation.

In this simulation, the laue optical branch gives the focusing-performance simulation and transports the focused signal through the Ge crystal mass.

For this purpose, the diffraction probability of photons by the crystal is no longer superposed onto photons as an efficiency parameter, but is defined as a discrete process at the Geant4 application layer [25, 26]. The process takes the diffraction probability $R(|\Delta\theta|)$ according to the angular deviation of the current photon relative to the crystal plane, and converts it into a finite equivalent diffraction mean free path λ_{diff} , so that the accumulated diffraction probability when the photon crosses a crystal path ℓ reproduces this value: $1 - \exp(-\ell/\lambda_{\text{diff}}) = R(|\Delta\theta|)$, namely $\lambda_{\text{diff}} = -\ell / \ln[1 - R(|\Delta\theta|)]$; to avoid double counting with standard photoelectric absorption, R is first normalized by the unabsorbed fraction ($R/(1 - A)$) and then converted into a mean free path. After this mean free path is returned to Geant4, the diffraction process and the standard electromagnetic processes (photoelectric absorption and Compton scattering) each sample interaction lengths in the same competition framework, and the shortest one decides which process occurs first, rather than forcibly intercepting photons at the crystal boundary. Photons not selected to undergo diffraction still continue transport according to the standard electromagnetic physics, and may be absorbed, scattered, or directly transmitted.

The diffraction probability is preferentially given by interpolation in an external XOP/CRYSTAL rocking-curve map as a function of angular deviation: this curve is calculated offline once in the preprocessing stage by the CRYSTAL/diff_pat module of the XOP toolkit (called through xoppylib, with DABAX crystallographic data as input) [27], and gives the Ge(111) reflectivity/transmissivity/absorption curves as a function of angular deviation; during transport, the angular deviation is calculated in real time for every step inside the crystal, and the current diffraction probability is obtained by interpolation on that curve, namely “offline library tabulation, probability used by angle during transport”. The mosaic orientation is implemented by applying a Gaussian angular perturbation to the ideal Bragg-plane normal, with the perturbation width given by the 30 arcsec mosaic FWHM; after diffraction occurs, the outgoing direction is mirror-reflected according to the perturbed plane normal, and the photon energy is unchanged. This split in which the “process class only handles the transport interface, and the independent model class handles probability and direction decisions” follows the idea of existing crystal Bragg-reflection extensions in Geant4 [28, 29], and extends it to the energy band needed here.

Here the reference signal is a monochromatic 511 keV point source whose rounded top-of-atmosphere flux scale ($F_0 = 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$, Section 1.1) is motivated by the dedicated INTEGRAL/IBIS compact-source upper limit [10], rather than by any single observed object. It is constructed as a MEGAlib far-field source [30], with the detailed construction given in Section 3.2.4. The signal is transported through the optics branch and injected into the detector mass model as a parallel-beam source; prompt cosmic-ray background and activation decays are transported directly in the detector/cryostat mass model. The primary background budget is therefore defined for the detector/cryostat branch, while the optics branch supplies the transported focal-plane signal phase space.

3.2 Source construction

After the focusing optics have been defined in Section 3.1, the calculation constructs the focused point source, the prompt atmospheric field, and the delayed radioactive population independently. Activation production and delayed-decay transport remain separate operations within the delayed chain. This preserves the normalization and particle–nuclide–position provenance of every component before the common detector response and event selection are applied.

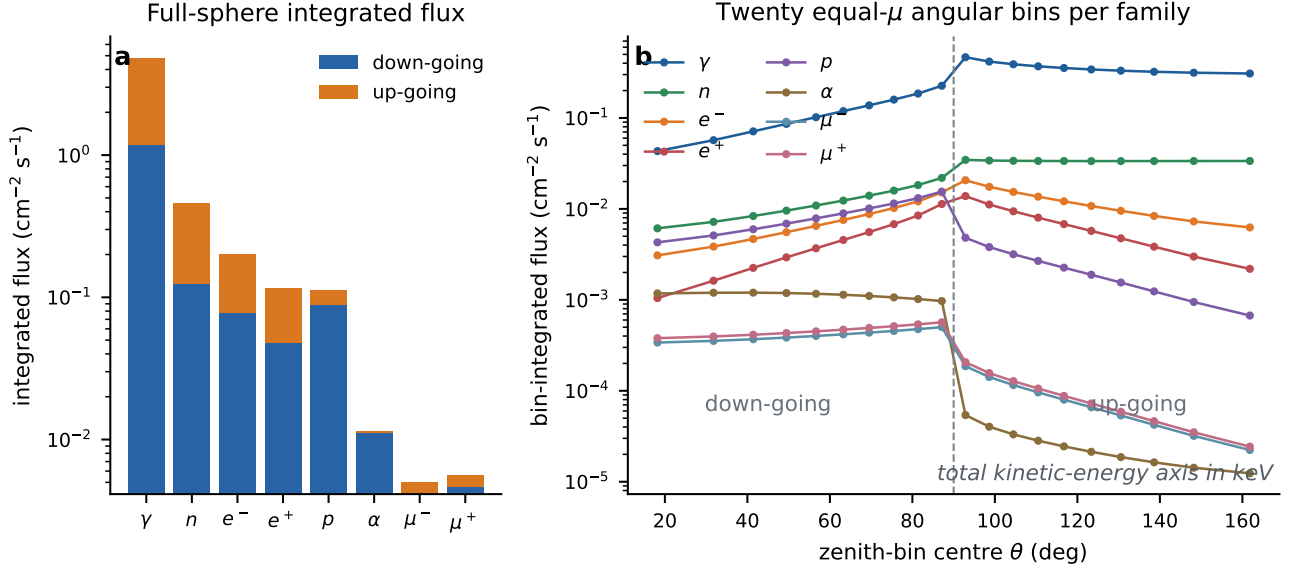


Figure 3: Corrected-total-kinetic-energy EXPACS/PARMA atmospheric field used for prompt transport and activation production. (a) Energy- and solid-angle-integrated fluxes of the eight particle families, separated into down-going and up-going hemispheres. (b) Bin-integrated fluxes in the 20 equal- μ zenith bins used by each far-field source. The environment is latitude 34° , longitude 100° , altitude 38 km, $R_c = 11.6$ GV, and $W = 118.3$.

3.2.1 Prompt atmospheric source

The prompt atmospheric field is derived from EXPACS/PARMA. PARMA3.0 is an analytical model parameterized from PHITS atmospheric-shower calculations and benchmarked against terrestrial cosmic-ray measurements, and EXPACS is its public tabulation and software interface [31, 32]. PHITS supplies the underlying particle-transport physics: primary cosmic rays interact with air nuclei, the secondary hadrons, leptons, photons, and neutrons are transported through the atmospheric column, and the resulting fluences are parameterized by altitude, geomagnetic cutoff, solar modulation, energy, and direction. In this work EXPACS/PARMA provides the differential incident flux as a function of particle species, kinetic energy, zenith angle, geographic position, altitude, cutoff rigidity, and solar activity, and is used only to construct the external radiation fields; detector response, vetoing, topology selection, and mission-time counting are applied downstream. The reference source definitions use latitude 34° , longitude 100° , altitude 38 km, vertical cutoff rigidity $R_c = 11.6$ GV, and the 2025-08-31 solar condition corresponding to $W = 118.3$.

The prompt source includes photons, neutrons, electrons, positrons, protons, alpha particles, and negative and positive muons. For each particle family the EXPACS/PARMA differential spectrum is converted into a MEGALib far-field area source covering the full zenith range $\theta = 0^\circ$ – 180° and full azimuth range $\phi = 0^\circ$ – 360° . The zenith dependence is represented by 20 differential equal- μ angular bins, where $\mu = \cos \theta$ and each bin subtends $\Delta\Omega = 0.6283185307$ sr: the first ten bins cover down-going particles, the last ten up-going or albedo-like particles. Both hemispheres are retained in the prompt transport and in the activation-production transport, so upward albedo particles are not discarded at source construction. Figure 3 shows the resulting full-sphere flux field.

The full-sphere integrated fluxes are 4.79966, 0.462239, 0.199002, 0.116981, 0.112301, 0.0114871, 0.00496893, and 0.00557001 $\text{cm}^{-2} \text{s}^{-1}$ for γ , n , e^- , e^+ , p , α , μ^- , and μ^+ , respectively. All families use 20 equal- μ angular bins and a 60 cm far-field start sphere. Minority families may be sampled more heavily, but every family retains its own transported exposure,

$$r_{\text{event},j} = \left(\sum_i \mathcal{T}_{ij} \right)^{-1}, \quad (2)$$

where $\mathcal{T}_{ij}$ is the exposure of subsample i for family j . Consequently, over-sampling changes Monte Carlo variance but not the expected physical rate.

The energy axis supplied to Cosima is total kinetic energy in keV: for non-alpha families $E_{\text{tot,keV}} = 1000E_{\text{MeV}}$, while for alpha particles $E_{\text{tot,keV}} = 4 \times 1000E_{\text{MeV/nucleon}}$. The retained source package contains 160 normalized spectra (20 per family) and 24 geometry-specific source cards. Their numerical integrals span 0.99999999991–1.000000000006, and all 480 spectrum references resolve to the corrected energy contract.

The angular and energy normalizations are deliberately factored. If $I_j(E, \mu)$ denotes the EXPACS/PARMA differential intensity for family j , the flux carried by equal- μ bin m is

$$\Phi_{j,m} = 2\pi \int_{\mu_m}^{\mu_{m+1}} d\mu \int I_j(E, \mu) dE, \quad \Delta\mu = 0.1, \quad (3)$$

so every bin subtends $\Delta\Omega = 2\pi\Delta\mu = 0.62831853$ sr. The first ten bins cover the down-going hemisphere and the second ten cover the up-going or albedo hemisphere; azimuth is complete in both cases. The spectrum within each bin is normalized to unit integral, while $\Phi_{j,m}$ carries the physical amplitude. This separation prevents a spectral-table normalization from being mistaken for a particle flux.

All eight families use the same 60 cm enclosing source sphere and the same directional convention. For a sampled direction, the source implementation draws the starting point on the corresponding projected disk and points the momentum into the enclosed mass model. Thus the geometrical rate factor is the projected area πR^2 , not the surface area $4\pi R^2$. Full-sphere in this context means complete incident solid angle; it does not mean uniform emission from arbitrary points on a spherical shell. The distinction is important for both prompt transport and activation production because an area error would propagate linearly into every derived rate and activity.

The corrected energy convention is likewise applied before any transport normalization. Electrons, positrons, photons, nucleons, and muons use the tabulated kinetic energy multiplied by 10^3 to convert MeV to keV. The alpha tables are per nucleon and therefore require the additional mass-number factor of four. Applying the same numerical conversion to alpha particles and the other seven families would shift their kinetic-energy distribution by another factor of four; treating the tabulated MeV values as keV would shift all families by three decades. The source contract records these two cases explicitly rather than relying on an implicit unit convention.

The retained package-level checks test the quantities that enter the physics normalization. There are 20 spectra for each of eight families, every spectrum is finite and non-negative, every numerical integral is unity to the precision quoted above, and every source-card reference resolves to the corrected-keV set. The geometry header of a source card must also agree with the mass model used by its transport. Prompt and activation-production inputs share the same field definition but remain separate calculations, so an activation product is never counted once as an immediate detector event and again as a later decay.

This construction also defines what can and cannot be inferred from the photon component. Its $4.79966 \text{ cm}^{-2} \text{ s}^{-1}$ integral is the broadband full-sphere photon flux across the retained energy domain. The table contains the annihilation enhancement and its adjacent continuum with their original energy dependence. The detector science-window rate is therefore obtained by transporting and response-convolving that broadband component; it is not the product of an independently fitted line flux and a detector efficiency. This avoids both double counting and an artificial separation of photons that are indistinguishable at the source-contract level.

3.2.2 Broadband atmospheric gamma component

The corrected atmospheric photon tables are treated as a single broadband gamma component. Their retained spectrum contains the annihilation feature around 511 keV together with the surrounding continuum. Adding a separate monoenergetic atmospheric source would therefore count the same physical feature twice. The gamma component instead follows the same 20-bin full-sphere geometry, exposure normalization, detector response, and trajectory-dependent family scaling as the other seven atmospheric families.

This contract preserves the distinction between the celestial signal and local atmospheric photons without introducing a second atmospheric-line normalization. The science signal remains the focused, on-axis EventList; atmospheric gamma events enter only through the broadband prompt catalogue and are decomposed by their incident-family tag after selection.

3.2.3 Delayed activation source

The delayed source is built from a separate activation-production transport. It is not counted as immediate detector background; instead, it records the isotope, material volume, incident family, and production position of each radioactive product. Ground-state half-lives are checked against NUBASE2020 [35]. For isotope k ,

$$A_k(t_{\text{flight}}) = P_k [1 - \exp(-\lambda_k t_{\text{flight}})], \quad \lambda_k = \frac{\ln 2}{t_{1/2,k}}, \quad (4)$$

where P_k is the production rate and the reference state is $t_{\text{flight}} = 15$ d. The resulting all-family day-15 activities are 1352.29749 Bq for SG3 and 211.387433 Bq for SH3.

An aggregate inventory alone is insufficient for this construction. The production calculation therefore retains a coordinate record at the moment a radioactive product is created, and the day-15 population is matched to those records by material volume, nuclide, and excitation state. The resulting source description is indexed by incident family, material category, production volume, and coordinate. These are source-construction quantities rather than detector counts: attenuation, decay transport, detector response, and event selection are applied only after the delayed particles are emitted from their recorded locations in the common geometry.

Position information is retained because downstream attenuation, active-shield probability, and topology depend on where a radionuclide was created. Production records are matched by volume, nuclide, excitation state, and coordinate before decay transport. The activity-normalized material distributions of incident families a and b may be compared through

$$D_{\text{TV}}(a, b) = \frac{1}{2} \sum_c |f_{a,c} - f_{b,c}|, \quad (5)$$

where $f_{a,c}$ is the fraction of family- a activity in material category c . This provenance is essential: a volume-integrated source could preserve total activity while erasing the family–material–nuclide correlations that determine the selected science-window background.

Let production row j carry activity weight w_j , with $A = \sum_j w_j$. Drawing M production positions with replacement using $p_j = w_j/A$ and assigning each point source flux A/M gives

$$\mathbb{E} \left[N_j \frac{A}{M} \right] = M p_j \frac{A}{M} = w_j, \quad (6)$$

so the sampled source is unbiased and conserves total activity. Each current family catalogue transports 10^6 delayed decays. The SH3 realization therefore contains 8×10^6 delayed triggers and retains 111 delayed records after the complete science-window, BGO, and topology selection. Its day-15 activities by incident family are 34.2836, 0.2663, 0.6754, 2.0320, 0.0684, 0.0040, 68.5316, and 105.5262 Bq for α , e^- , e^+ , γ , μ^- , μ^+ , n , and p .

Table 1 summarizes the source layers. Source construction ends before detector response, anticoincidence, and topology selection; the latter operations are defined in Section 3.4.

The selected delayed events retain their source-parent identity and exact production volume. In SG3, the leading groups include ^{62}Cu in an L2 copper support ring and in the 50 mK mixing-chamber plate. In SH3, the largest group is ^{62}Cu in an L5 copper ring, followed by ^{62}Cu in the TES heat-sink ring; tungsten-frame and aluminium-shell groups are smaller. These selected-event origins, rather than the unfiltered total inventory alone, drive the geometry interpretation in Section 4.1.

3.2.4 Reference point-source construction

The focused signal stream is defined by a monochromatic 511 keV reference point source. Its flux scale is set not by any single observed object but by a rounded comparison to the dedicated INTEGRAL/IBIS compact-source limit: $F_0 = 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$ at the top of the atmosphere (Section 1.1) [10].

Because such a source lies at astronomical distance, it is modelled as an ideal on-axis point: in the optics calculation it is a monochromatic 511 keV far-field beam whose photons arrive parallel to the optical axis and are sampled uniformly over the incident aperture A_{inc} of Equation (1). No finite angular size is assigned to

Table 1: Source layers used by both geometries. The table defines source construction; detector response and selection are applied downstream.

Layer	Physical role	Current implementation
Corrected atmospheric field	Prompt detector hits and activation production	Eight families, 20 equal- μ bins per family, full sphere, 60 cm start radius; correct total kinetic energy in keV.
Broadband gamma	Atmospheric photons including the annihilation feature	One retained broadband component; no additive monoenergetic normalization.
Activation production	Family–nuclide–volume–position history	The same corrected atmospheric field, with NUBASE-checked ground-state decay data.
Day-15 radioactive population	Flight-time activity used for delayed transport	SG3: 1352.29749 Bq; SH3: 211.387433 Bq.
Production-position decay	Delayed particles emitted from recorded material coordinates	10^6 decays per family; eight independently normalized family catalogues per geometry.

the source itself. The angular structure at the focal plane is produced by the 30 arcsec mosaic spread, the XOP/CRYSTAL rocking curve, and the 10 m focusing geometry, so the accepted photons form a finite spot rather than a single ray. Three independent optics samples totalling 1.5×10^5 incident photons yield 37,256 diffracted focal-plane crossings; 37,194 of them fall inside the $r = 1.898$ cm beryllium entrance window. Their spot radii are $r_{90} = 1.03$ cm and $r_{99} = 1.25$ cm. The recorded position and direction of each accepted crossing form the common detector-side EventList used for both geometries.

The focused rate at the detector entrance is

$$R_{fp} = F_{TOA} A_{eff}(511 \text{ keV}) T_{atm,45}. \quad (7)$$

Equivalently, the generating optics sample represents an exposure $N_{fp}/[F_{TOA} A_{eff} T_{atm,45}]$, and every EventList entry carries the inverse of that exposure as its physical rate weight. This separates detector-side acceptance from the top-of-atmosphere flux normalization and applies the latter only once. Because the optics calculation contains source photons only, all backgrounds still enter exclusively through the atmospheric and delayed-activation streams defined above. Here $T_{atm,45} = 0.652034$ at day 15. In SG3, 28,612 entries lie in the measured science window and 28,006 pass the final topology selection, giving $A_{eff} = 15.12324 \pm 0.04492 \text{ cm}^2$. In SH3 the corresponding counts are 28,434 and 27,936, giving $15.08544 \pm 0.04503 \text{ cm}^2$. After atmospheric transmission and the day-15 accidental-coincidence survival are applied, the rates at $F_{TOA} = 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$ are $9.5362 \times 10^{-4} \text{ s}^{-1}$ for SG3 and $9.7904 \times 10^{-4} \text{ s}^{-1}$ for SH3. Thus the chimney changes the selected area by only 0.25% while substantially changing the background environment.

The effective area quoted above is the on-axis value, representing best-case pointing. A point source offset from the optical axis, or residual attitude jitter during flight, reduces and shifts the focused response through the finite angular acceptance (field of view) of the Laue lens [11, 13]; this off-axis dependence is treated as a pointing systematic on the signal effective area rather than as a change to the on-axis reference construction. The current side-entry response is a fixed 45° -elevation reference, not a simulated low-elevation Galactic-centre observation.

3.3 Background-led geometry design and statistical treatment

The design starts from events that reach the science window. For SG3, final prompt events are grouped by incident family and delayed events by incident family, source-parent nuclide, production material, volume, and coordinate. This component-first logic follows detailed COSI and DIXE background studies [17, 18]. It shows that the staged cold region lies directly over the SG3 focal plane. SH3 therefore translates the TES laterally into the side chimney while retaining the optical acceptance and surrounding the new focal region with active BGO.

Cosima deposits are aggregated by TES pixel before an independent Gaussian response with $\Delta E_{FWHM} = 0.420 \text{ keV}$ ($\sigma = 0.17836 \text{ keV}$) is applied. Measured hits below 0.300 keV are removed, and the remaining hits are summed before the science-window, BGO, and topology tests. All geometry comparisons use this identical response contract.

For stream k and stage j , the physical rate, Monte Carlo standard error, and effective support are

$$\hat{R}_{k,j} = \sum_{i \in (k,j)} w_i, \quad \sigma_{k,j} = \left(\sum_{i \in (k,j)} w_i^2 \right)^{1/2}, \quad N_{\text{eff},k,j} = \frac{\hat{R}_{k,j}^2}{\sigma_{k,j}^2}. \quad (8)$$

Counts report transported support, whereas weighted rates report the physical background. We also retain the stage survival $q_{k,j} = \hat{R}_{k,j} / \hat{R}_{k,\text{window}}$ and the final component share $f_k = \hat{R}_{k,\text{final}} / \sum_b \hat{R}_{b,\text{final}}$, because a small transported count can still carry an important physical rate when component exposures differ.

Within an independently normalized transported component the event weights are constant, so a two-sided 95% Garwood interval for the Poisson count can be multiplied by the component weight to give an exact rate interval. A zero selected count with defined exposure therefore gives an upper limit rather than a claim of zero physical background. Focused-signal acceptance is binomial and is checked with the corresponding exact interval. These componentwise bounds diagnose finite transport support; they are not a joint-coverage band and do not include radiation field, cross-section, calibration, or pointing systematics. The primary mission uncertainty instead propagates the background-transport, common-time replay, selected-area, and accidental-probe terms by the delta method.

For trajectory bin ℓ , live fraction L_ℓ , and duration Δt_ℓ ,

$$N_s(T) = \sum_{\ell \leq T} S_\ell L_\ell \Delta t_\ell, \quad N_b(T) = \sum_{\ell \leq T} B_\ell L_\ell \Delta t_\ell. \quad (9)$$

We report both the Gaussian counting measure $Z_G = N_s / \sqrt{N_b}$ and the Asimov form

$$Z_A = \sqrt{2 \left[(N_s + N_b) \ln \left(1 + \frac{N_s}{N_b} \right) - N_s \right]}. \quad (10)$$

The minimum flux is obtained from the same signal kernel by solving for $Z = 3$ or 5 . This keeps the mission comparison on one statistical and physical normalization.

3.4 Detector event selection and veto definitions

The focused-signal, prompt-atmospheric, and delayed-activation catalogues are placed on one explicit time axis. The same measured science window, active BGO anticoincidence, and topology/field-of-view definition is then applied to both geometries. Prompt gamma already carries the retained atmospheric annihilation feature, so the rate budget contains no additional atmospheric catalogue.

3.4.1 Time axis and rate normalization

Each stream enters with a per-record rate weight r_{km} , with total $R_k = \sum_m r_{km}$. For a reference interval T , independent Poisson draws

$$N_k \sim \text{Pois}(R_k T) \quad (11)$$

are sampled with probabilities r_{km}/R_k , assigned uniform times in $[0, T]$, merged, and ordered. Records separated by at most $\tau = 1 \mu\text{s}$ form one coincidence group. Five 2×10^5 s anchor realizations at days 0, 5, 10, 15, and 20 test this construction. Figure 4 compares the common-time rate with the independently normalized direct rate, while Figure 5 shows the weighted pre-veto spectra of the two geometries. At day 15 the common-time/direct ratios are 0.9665 for SG3 and 1.0172 for SH3, consistent with the finite anchor counts; the corresponding signal accidental-survival factors are 0.967071 and 0.995341.

The mission fold uses the common-time expectation in each trajectory bin,

$$L_\ell = \exp \left\{ -\tau \left[R_{\text{prompt},\ell}^{\text{occ}} + R_{\text{delayed},\ell}^{\text{occ}} \right] \right\}. \quad (12)$$

The weak focused source is excluded from the exponent; a brighter-source application would add its occupancy explicitly. Atmospheric and activation rates vary on mission-day scales, whereas grouping acts on microseconds, so each time bin is treated as a stationary superposition. The five explicit anchors validate the coincidence construction; the 81-node fold uses its expectation rather than drawing a new stochastic realization in every bin.

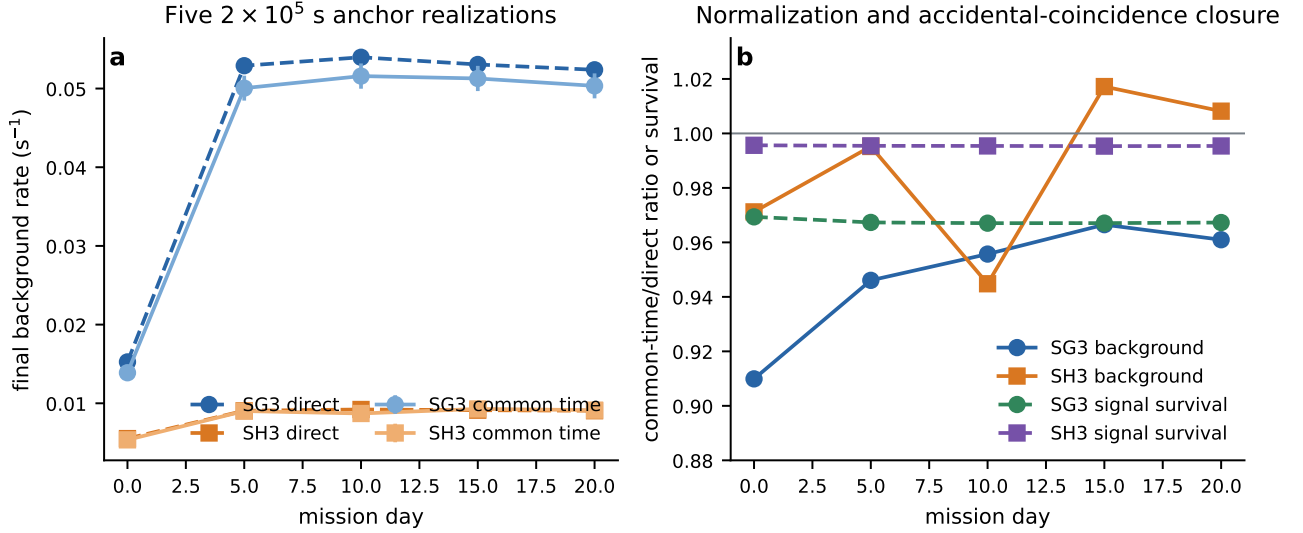


Figure 4: Common-time normalization at five mission anchors. (a) Direct no-coincidence and 2×10^5 s common-time final background rates for SG3 and SH3; error bars are Poisson standard errors of the anchor realization. (b) Common-time/direct background ratios and focused-signal accidental-survival factors. The same $1 \mu\text{s}$ grouping contract is used at all anchors.

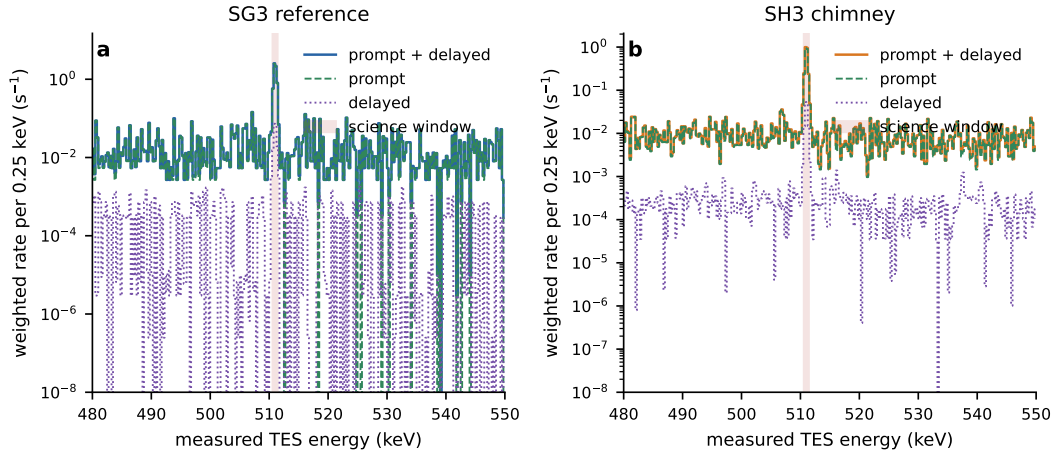


Figure 5: Response-convolved, weighted prompt-plus-delayed spectra before active vetoes for (a) the SG3 reference geometry and (b) the SH3 chimney geometry. The shaded band marks the 510.58–511.42 keV science window; absolute cut-flow rates are reported in Tables 2 and 5.

3.4.2 Active anticoincidence gate

The first layer is active BGO anticoincidence. For coincidence group c ,

$$E_{\text{TES}}^{(c)} = \sum_{j \in c} E_{\text{TES},j}, \quad E_{\text{sh}}^{(c)} = \sum_{j \in c} E_{\text{sh},j}. \quad (13)$$

The group enters the science window when

$$W_{511} = [510.58, 511.42] \text{ keV} \quad (14)$$

and passes the active shield when

$$E_{\text{sh}}^{(c)} < E_{\text{veto}}, \quad E_{\text{veto}} = 50 \text{ keV}. \quad (15)$$

In the SG3 direct catalogue, the gate changes the prompt sample from 1083 records and 5.79018 s^{-1} to seven records and $1.87041 \times 10^{-2} \text{ s}^{-1}$, and the delayed sample from 1295 records and 0.169494 s^{-1} to 436 records and

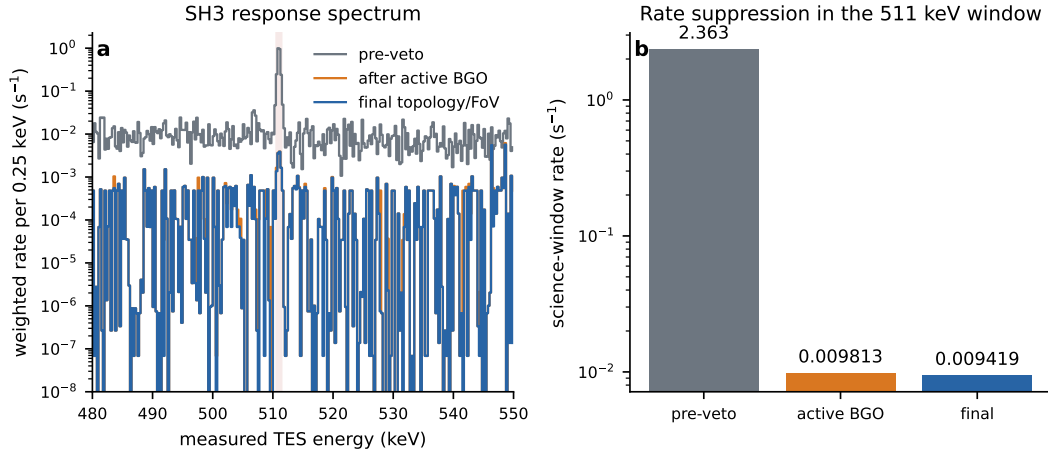


Figure 6: Active-BGO suppression in the SH3 chimney geometry. (a) Weighted 480–550 keV spectra before anticoincidence, after active BGO, and after the final topology/field-of-view selection. (b) Corresponding rates in the science window; active BGO reduces the direct background from 2.36263 to $9.8133 \times 10^{-3} \text{ s}^{-1}$.

$4.34116 \times 10^{-2} \text{ s}^{-1}$. All 28,612 signal records in the measured window pass active anticoincidence. Figure 6 shows the still stronger SH3 BGO suppression.

The quoted flux thresholds apply to a line whose intrinsic width, after detector response, remains small compared with this window; a substantially broadened source line would require convolving the source line profile with the response and re-optimizing the window.

3.4.3 Compton topology and field-of-view consistency

The second layer applies a Compton-kinematic topology and field-of-view consistency test to the TES hits that survive the active-shield gate; its physical basis is Compton-camera cone reconstruction [30]. An incident photon of energy E_0 Compton-scatters at a first interaction point $\mathbf{r}_1$, depositing E_1 , and the scattered photon of energy $E_{\text{sc}} = E_0 - E_1$ propagates to a second interaction point $\mathbf{r}_2$; the scattering angle follows from

$$\cos \theta_C = 1 - m_e c^2 \left(\frac{1}{E_{\text{sc}}} - \frac{1}{E_1 + E_{\text{sc}}} \right). \quad (16)$$

Given the scattered-photon direction $\hat{\mathbf{u}}_{12} = (\mathbf{r}_2 - \mathbf{r}_1)/|\mathbf{r}_2 - \mathbf{r}_1|$, the source direction is constrained to a reconstruction cone with apex $\mathbf{r}_1$, axis $-\hat{\mathbf{u}}_{12}$, and half-angle θ_C , whose projection on the sky is the event circle. We use this first-scatter cone as an aperture-consistency test: the event is retained when the cone intersects the side Be-window disk (radius 1.898 cm), i.e. whether the reconstructed arrival direction is consistent with entry through the aperture.

Because the interaction position within a single TES pixel ($1.5 \times 1.5 \times 3.0 \text{ mm}$) is unresolved, each hit is represented not by its centroid alone but by $8 + 1 = 9$ representative points—the eight pixel corners and the centroid—carrying the sub-pixel position uncertainty explicitly into the geometric reconstruction. A two-hit event in a given scatter order therefore enumerates $9 \times 9 = 81$ first/second position combinations and constructs one reconstruction cone per combination (each cone sampled at 24 azimuths, propagated to the Be-window plane, and tested against the disk by both its sampled points and the connecting segments); the order is deemed field-of-view consistent if any of these 81 trajectories yields a cone that intersects the disk. This is a deliberately permissive consistency test: an event is retained if any sub-pixel interaction position is consistent with entry through the aperture. The layer consequently retains about 95% of the in-window multi-hit events and removes only about 10–14% of the background, consistent with its role as a loose consistency veto rather than an imaging reconstruction.

The three event classes are handled accordingly. Single-pixel events are retained as calorimetric candidates. For two hits, both interaction orders and the representative sub-pixel positions are tested, and either consistent

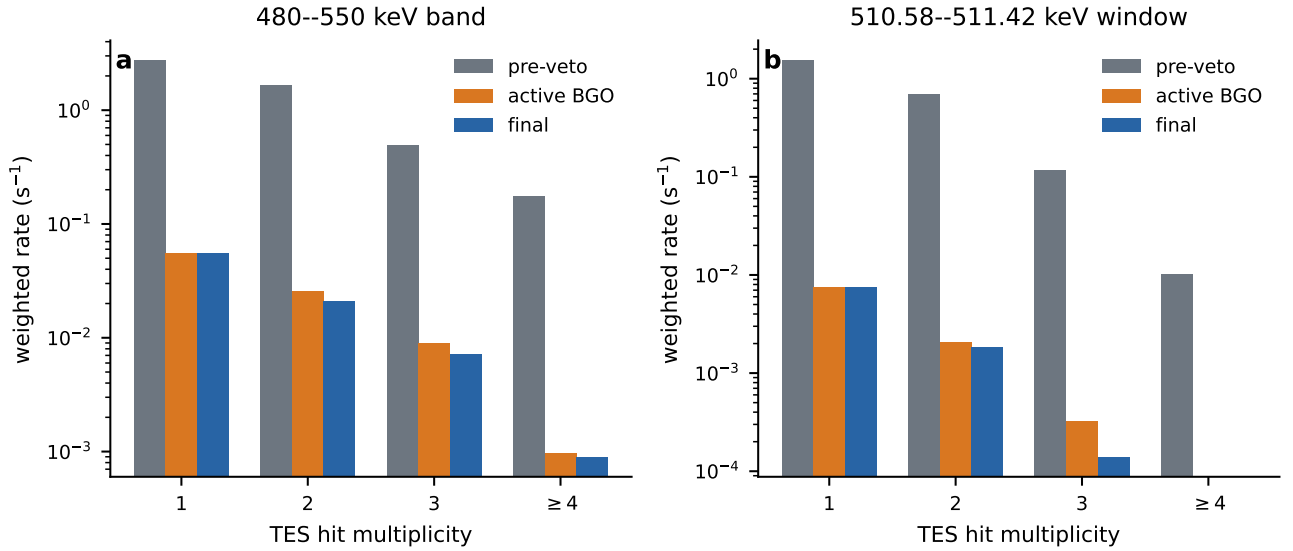


Figure 7: SH3 weighted background rate by TES hit multiplicity before active veto, after active BGO, and after topology/field-of-view selection in (a) the 480–550 keV band and (b) the 510.58–511.42 keV science window. Multi-hit events are retained when their Compton cone is consistent with the entrance aperture.

Table 2: SG3 reference-geometry science-window cut-flow. Background entries are direct no-coincidence counts/rates; signal entries give counts/effective area for the common 37,194-ray focal sample.

Stream	Pre-veto	After active veto	Final topology/FoV
Prompt background	1083 / 5.79018 s ⁻¹	7 / 1.87041 × 10 ⁻² s ⁻¹	6 / 1.60321 × 10 ⁻² s ⁻¹
Delayed activation	1295 / 0.169494 s ⁻¹	436 / 4.34116 × 10 ⁻² s ⁻¹	394 / 3.82661 × 10 ⁻² s ⁻¹
Total background	2378 / 5.95968 s ⁻¹	443 / 6.21157 × 10 ⁻² s ⁻¹	400 / 5.42982 × 10 ⁻² s ⁻¹
Focused signal	28612 / 15.45048 cm ²	28612 / 15.45048 cm ²	28006 / 15.12324 cm ²

order is sufficient. For three to six hits, all orders are ranked by the Compton sequence residual

$$Q = \sum_{i=1}^{n-2} (\cos \theta_{C,i} - \hat{\mathbf{u}}_{i-1,i} \cdot \hat{\mathbf{u}}_{i,i+1})^2, \quad (17)$$

and the best order is subjected to the same cone–aperture test. The SG3 topology layer retains 400 of 443 active-veto background records, with a weighted-rate survival of 87.41%, and 28,006 of 28,612 signal records. In SH3 it retains 123 of 134 background records, with 95.98% weighted-rate survival, and 27,936 of 28,434 signal records. Figure 7 resolves these samples by TES hit multiplicity.

Unreconstructed events remain in the rate accounting but are not labelled as field-of-view-consistent candidates. The measured multiplicity distributions show why a permissive policy is appropriate: active BGO supplies the large rate reduction, whereas topology removes a smaller residual population while preserving the focused multi-hit signal. Table 2 gives the SG3 reference cut-flow on the direct, no-coincidence normalization; its mission rate is obtained from the common-time fold.

The retained hit-level catalogues provide an internal check of this classification without another transport. For each selected event they preserve measured hit energy, TES layer, pixel position, and multiplicity. Re-evaluating the broad and science windows from these fields reproduces the tabulated stage counts and confirms that the topology layer is applied after, rather than folded into, the active-BGO decision.

3.5 Mission-time trajectory and analytic rate folding

MeV balloon observations are background dominated: atmospheric secondaries and cosmic-ray-induced activation set the rate floor, and both respond to the flight environment [17, 36]. Detector-coupled Monte Carlo through a fixed geometry provides the selected day-15 signal and background rates under a common response, veto, and energy window, but those rates describe only one longitude, latitude, and altitude. During a multi-day

float the residual atmospheric depth, geomagnetic cutoff, and activation inventory evolve, so a constant-rate extrapolation of the day-15 snapshot is not a substitute for a trajectory fold. We therefore map the reference rates onto a continuous 20 d performance estimate and retain explicit common-time anchors as an independent check of the time-axis construction.

The adopted trajectory is a *synthetic reference profile*, not balloon telemetry and not a source-visibility schedule. It contains 81 nodes from day 0 to day 20 at 0.25 d spacing, with a mild mid-latitude float envelope around 38.75 km altitude, 34° latitude, and 100° longitude. Altitude is retained because changes in residual atmospheric depth affect both the particle spectra that drive prompt background and activation and the 511 keV atmospheric transmission multiplying the focused signal. Latitude and longitude supply a controlled reference modulation of the geomagnetic environment; they do not encode Earth occultation, repointing losses, or the visibility of a named source. The fold is therefore a reference-exposure sensitivity calculation rather than a forecast for a particular launch campaign.

The fold contains five physically separate layers. First, for a monochromatic point source,

$$R_{\text{sig}}(t) = F_{511} A_{\text{eff}} T_{\text{atm},511}(t) \varepsilon_{\text{det}}, \quad (18)$$

where the fixed-45° slant transmission is $T_{\text{atm},511}(t) = \exp[-\mu_{\text{air}} X(t) / \cos 45^\circ]$ and $T_{15,45} = 0.652034$. Second, the selected prompt rate is

$$R_{\text{prompt},j}(t) = \sum_b \Phi_b(t) K_{j,b}, \quad (19)$$

where each of the eight family fluxes $\Phi_b(t)$ has its own 81-node scaling and $K_{j,b}$ is fixed by the geometry-specific detector catalogue. The retained broadband gamma family includes its annihilation feature in this same sum.

Third, activation production for family a and source-parent nuclide k is

$$P_{a,k}(t) = \Phi_a(t) Y_{a,k}, \quad (20)$$

using the SG3 or SH3 day-15 inventory appropriate to that geometry. Fourth, the inventory is advanced by

$$N_{a,k}(t + \Delta t) = N_{a,k}(t) e^{-\lambda_k \Delta t} + \frac{P_{a,k}(t)}{\lambda_k} (1 - e^{-\lambda_k \Delta t}), \quad A_{a,k}(t) = \lambda_k N_{a,k}(t), \quad (21)$$

and, fifth, the selected delayed rate is

$$R_{\text{delayed},j}(t) = \sum_a \sum_k A_{a,k}(t) \varepsilon_{j,a,k}. \quad (22)$$

These equations keep celestial attenuation, local prompt modulation, activation production, and radioactive decay distinct.

The five common-time anchor realizations at days 0, 5, 10, 15, and 20 each span 2×10^5 s and validate the mapping from direct catalogues to the trajectory axis. The final calculation applies the corresponding family scales and inventory evolution at all 81 nodes, then evaluates the cumulative counts in Equation (9). Both SG3 and SH3 therefore use matched sources, response, selection, timing, and signal normalization.

The five anchor calculations provide a quantitative closure test rather than a visual comparison alone. Across days 0, 5, 10, 15, and 20, the SG3 common-time/direct background ratios are 0.9099, 0.9461, 0.9557, 0.9665, and 0.9610. The corresponding SH3 ratios are 0.9712, 0.9952, 0.9448, 1.0172, and 1.0082. Their deviations from unity track the finite number of final-window events in each 2×10^5 s realization: 293–1048 for SG3 and 1071–1856 for SH3. No empirical correction is fitted to these ratios. The 81-node fold uses the Poisson expectation, while the anchors test that explicit event grouping converges to that expectation within its counting support.

The same anchors separately probe accidental loss of a weak focused event. Signal-survival factors span 0.96705–0.96937 in SG3 and 0.99534–0.99564 in SH3. The higher SH3 survival is a consequence of its lower detector occupancy, not a change in the signal selection. At day 15 the survival values are 0.967071 and 0.995341; they are multiplied after the common 37,194-ray acceptance calculation. This ordering keeps geometrical acceptance, atmospheric transmission, and random-coincidence loss as independently auditable factors.

For each trajectory node the prompt response coefficient of every incident family remains fixed and only its external field scale changes. The delayed calculation is different: production is rescaled by family first, each source-parent inventory is advanced with its own half-life, and only then is its geometry-specific selected-decay efficiency applied. A single scale factor on the total day-15 delayed rate would erase short- versus long-lived behaviour and would not reproduce the initial rise of the mission curve. The recurrence in Equation (21) is therefore evaluated separately for every retained family–parent component at all 81 nodes.

Threshold-crossing times are obtained from the cumulative signal and background kernels on this same grid, using interpolation only between adjacent trajectory nodes. The statistical error on the reported Gaussian 3σ flux combines independent transport-background, timeline, selected-area, and accidental-probe terms by the delta method. Its relative 1σ value is 7.42% for SG3 and 9.36% for SH3. The latter contains an 18.67% relative uncertainty on the selected background itself, which enters a background-dominated flux threshold with approximately one-half weight. These terms describe Monte Carlo and common-time support; they are not a substitute for radiation-field, cross-section, calibration, or pointing systematics.

4 Results

4.1 What sets the 511 keV line-window background?

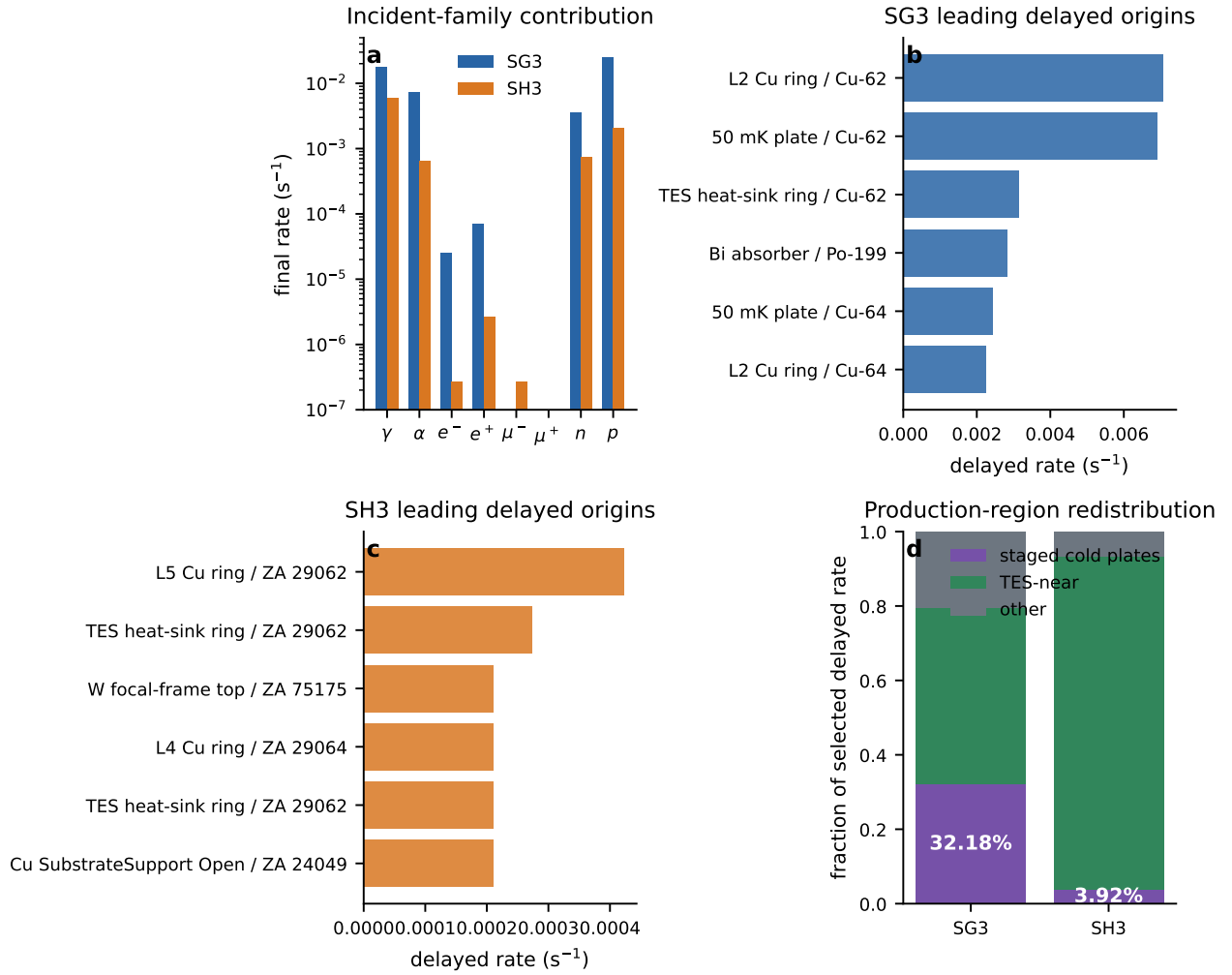
The SG3 reference catalogue provides the selected-event diagnosis. In the 510.58–511.42 keV window, direct no-coincidence selection retains six prompt and 394 delayed records, corresponding to 1.60321×10^{-2} and $3.82661 \times 10^{-2} \text{ s}^{-1}$. Their total direct rate is $5.42982 \times 10^{-2} \text{ s}^{-1}$; the independently realized day-15 common-time rate is $5.12802 \times 10^{-2} \text{ s}^{-1}$. Table 3 resolves the direct rate by incident family.

The family decomposition changes the physical interpretation of the reference background. Prompt leakage is supported only by the broadband gamma sample, whereas proton-induced delayed activity is the largest direct component. The leading SG3 delayed groups are ^{62}Cu in the L2 copper support ring and the 50 mK mixing-chamber plate, followed by the TES heat-sink ring and the bismuth absorber. Thus total inventory alone is insufficient: the relevant quantity is the activity that is both produced in a geometrically coupled location and selected in the science window.

The production-region view makes the geometry lever explicit. In SG3, 32.18% of the selected delayed rate originates in the DR/mixing-chamber and staged cold plates and another 47.29% in TES-near structures. The SH3 translation removes the focal plane from the direct cold-stage projection; the corresponding staged-cold-region fraction is 3.92%, while the selected delayed rate falls from 3.82661×10^{-2} to $3.62467 \times 10^{-3} \text{ s}^{-1}$.

Table 3: SG3 final direct background in the science window. Counts and rates are shown together because independently normalized families carry different event weights.

Component	Records	Rate / s^{-1}	Total / %	Selected-event origin
Prompt γ	6	1.60321×10^{-2}	29.526	broadband atmospheric photons
Delayed α	26	7.47607×10^{-3}	13.769	activated near-field materials
Delayed e^{-}	38	2.53793×10^{-5}	0.047	activated materials
Delayed e^{+}	24	7.08887×10^{-5}	0.131	activated materials
Delayed γ	260	1.71400×10^{-3}	3.157	photon-induced activity
Delayed n	11	3.63023×10^{-3}	6.686	neutron-induced activity
Delayed p	35	2.53496×10^{-2}	46.686	proton-induced activity
Total	400	5.42982×10^{-2}	100.000	prompt plus delayed

**Figure 8:** Selected background origins in the matched SG3 and SH3 catalogues. (a) Final direct rate by incident family. (b,c) Leading SG3 and SH3 delayed source-volume/parent-nuclide groups. (d) Selected delayed rate grouped into staged cold plates, TES-near structures, and other structures. Moving the focal plane into the chimney reduces the staged-cold-region share from 32.18% to 3.92%.

4.2 Final side-chimney active-shield geometry

The final SH3 geometry places the six TES layers along the lateral chimney axis at $z' = -2.80$ cm. A 3 mm aluminium wall joins the chimney continuously to the DR; the chimney bottom remains 0.25 cm above the DR bottom. A compact tungsten frame supports the focal plane, and three active BGO volumes cover the chimney side, the front optical annulus, and the rear cold-port annulus. Figure 9 connects this layout to the cut-flow and final composition, while Table 4 records the dimensions used by the transport.

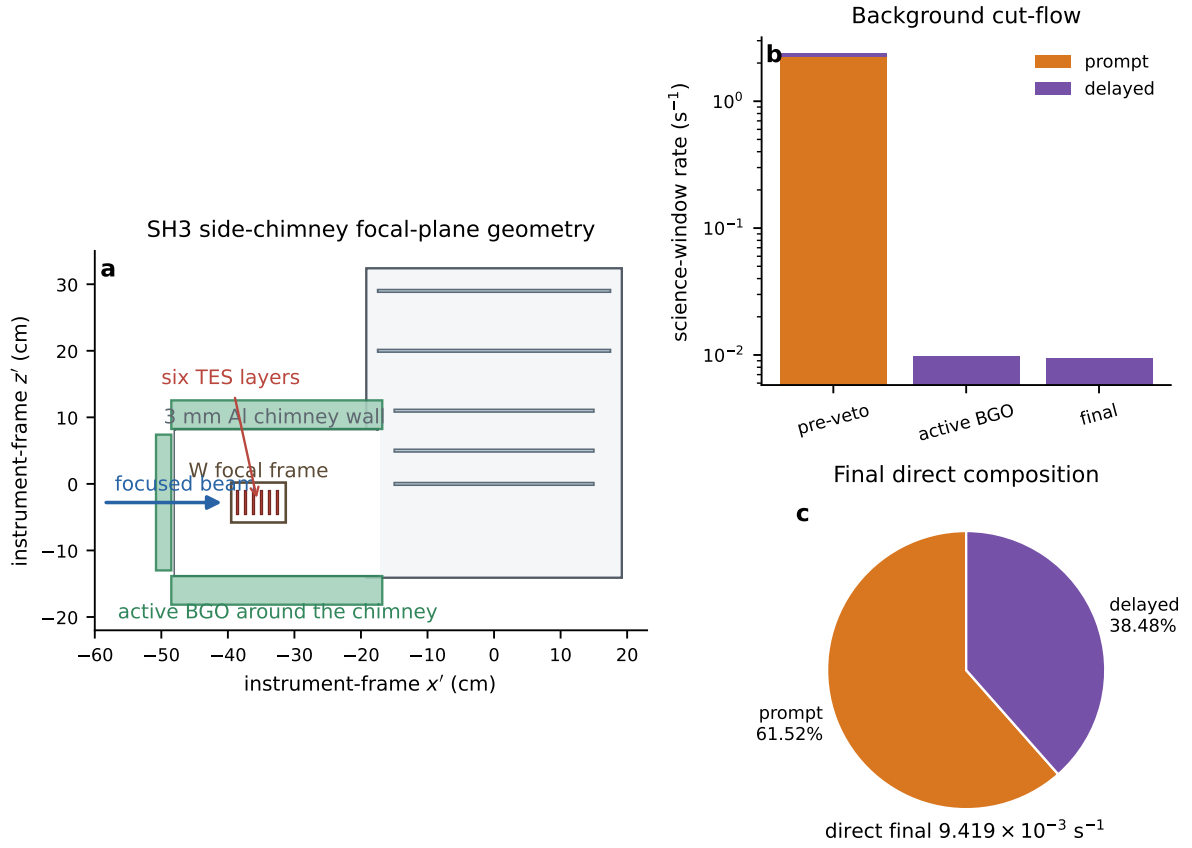


Figure 9: SH3 chimney geometry and selected background. (a) Instrument-frame section showing the lateral aluminium chimney, six TES layers, tungsten focal frame, focused beam, staged cold plates, and active BGO. (b) Prompt and delayed direct rates through the science-window, active-BGO, and final-topology stages. (c) Composition of the final direct background ($9.41868 \times 10^{-3} \text{ s}^{-1}$).

Table 4: Final SH3 side-chimney geometry in local instrument coordinates.

Element	Final setting	Physical role
Aluminium chimney	inner/outer radius 10.75/11.05 cm; 3 mm wall	laterally separates the TES from the staged cold region
Common axis and clearance	$z' = -2.80 \text{ cm}$; chimney/DR bottoms $-13.85/ -14.10 \text{ cm}$	continuous welded interface with 0.25 cm clearance
TES focal plane	six Ta-pixel layers along the chimney axis	preserves the detector response and multilayer stopping depth
Tungsten focal frame	outer/inner square 6.00/5.40 cm; depth 2.00 cm	compact local support and aperture definition
Cold-stage/optical ports	1.50 cm cold ports; 6.00 cm upper optical cut	retains thermal services and focused-beam access
Active BGO	side shield, front optical annulus, rear cold-port annulus	event-by-event anticoincidence around the chimney
BGO threshold	50 keV offline deposited energy; 80 keV native trigger reference	common active-veto decision

The selected-area comparison verifies that this displacement does not trade away the optical response. SG3 gives $15.12324 \pm 0.04492 \text{ cm}^2$ and SH3 gives $15.08544 \pm 0.04503 \text{ cm}^2$ after final selection, so the chimney retains 99.75% of the reference effective area.

4.3 Full-chain background and statistical support

Table 5 follows the SH3 streams through the same response and event sequence. The background rates in this table are direct no-coincidence values; the mission calculation applies the measured common-time factor at

every trajectory node.

Table 5: SH3 response-convolved science-window cut-flow. Background entries are direct no-coincidence counts/rates; signal entries give counts/effective area.

Stream	Pre-veto	After active BGO	Final topology/FoV
Prompt background	3212 / 2.23515 s^{-1}	$12 / 5.79401 \times 10^{-3} \text{ s}^{-1}$	$12 / 5.79401 \times 10^{-3} \text{ s}^{-1}$
Delayed activation	3249 / 0.127481 s^{-1}	$122 / 4.01926 \times 10^{-3} \text{ s}^{-1}$	$111 / 3.62467 \times 10^{-3} \text{ s}^{-1}$
Total background	6461 / 2.36263 s^{-1}	$134 / 9.81327 \times 10^{-3} \text{ s}^{-1}$	$123 / 9.41868 \times 10^{-3} \text{ s}^{-1}$
Focused signal	28434 / 15.35436 cm^2	28434 / 15.35436 cm^2	27936 / 15.08544 cm^2

Active BGO reduces the total direct rate to 0.4154% of its pre-veto value. The topology test then retains 95.98% of the BGO-selected rate. Prompt gamma contributes $5.79401 \times 10^{-3} \text{ s}^{-1}$ (61.52%) and delayed activation $3.62467 \times 10^{-3} \text{ s}^{-1}$ (38.48%). The day-15 common-time background is $9.280 \times 10^{-3} \text{ s}^{-1}$, which is the value used for the primary mission projection.

Table 6: Finite-transport support of the SH3 final direct background. Intervals are two-sided 95% Garwood count intervals scaled by the component event weight; subtotal upper endpoints are diagnostics, not joint confidence intervals.

Component	Records	Weight / s^{-1}	Central rate / s^{-1}	95% interval / s^{-1}	Total / %
Prompt α	0	5.4193×10^{-3}	0	$[0, 1.9991 \times 10^{-2}]$	0
Prompt e^-	0	1.8840×10^{-3}	0	$[0, 6.9500 \times 10^{-3}]$	0
Prompt e^+	0	5.4388×10^{-3}	0	$[0, 2.0063 \times 10^{-2}]$	0
Prompt γ	12	4.8283×10^{-4}	5.7940×10^{-3}	$[2.9939 \times 10^{-3}, 1.0121 \times 10^{-2}]$	61.516
Prompt μ^-	0	1.3461×10^{-3}	0	$[0, 4.9655 \times 10^{-3}]$	0
Prompt μ^+	0	3.9779×10^{-3}	0	$[0, 1.4674 \times 10^{-2}]$	0
Prompt n	0	5.5440×10^{-3}	0	$[0, 2.0451 \times 10^{-2}]$	0
Prompt p	0	1.2360×10^{-2}	0	$[0, 4.5595 \times 10^{-2}]$	0
Prompt subtotal	12	–	5.7940×10^{-3}	upper sum 1.4281×10^{-1}	61.516
Delayed α	19	3.4284×10^{-5}	6.5139×10^{-4}	$[3.9218 \times 10^{-4}, 1.0172 \times 10^{-3}]$	6.916
Delayed e^-	1	2.6633×10^{-7}	2.6633×10^{-7}	$[6.7429 \times 10^{-9}, 1.4839 \times 10^{-6}]$	0.003
Delayed e^+	4	6.7536×10^{-7}	2.7015×10^{-6}	$[7.3606 \times 10^{-7}, 6.9168 \times 10^{-6}]$	0.029
Delayed γ	52	2.0320×10^{-6}	1.0567×10^{-4}	$[7.8917 \times 10^{-5}, 1.3857 \times 10^{-4}]$	1.122
Delayed μ^-	4	6.8409×10^{-8}	2.7364×10^{-7}	$[7.4557 \times 10^{-8}, 7.0062 \times 10^{-7}]$	0.003
Delayed μ^+	0	3.9662×10^{-9}	0	$[0, 1.4631 \times 10^{-8}]$	0
Delayed n	11	6.8532×10^{-5}	7.5385×10^{-4}	$[3.7632 \times 10^{-4}, 1.3488 \times 10^{-3}]$	8.004
Delayed p	20	1.0553×10^{-4}	2.1105×10^{-3}	$[1.2892 \times 10^{-3}, 3.2595 \times 10^{-3}]$	22.408
Delayed subtotal	111	–	3.6247×10^{-3}	upper sum 5.7733×10^{-3}	38.484
Total background	123	–	9.4187×10^{-3}	component upper sum 1.4858×10^{-1}	100.000

The component table exposes where finite transport support remains sparse, but the primary flux-threshold uncertainty is not obtained by adding its interval endpoints. The matched delta-method budget gives an 18.67% relative background transport term, a 1.13% timeline term, a 0.30% selected-area term, and a negligible accidental-probe term. Their propagation gives a 9.36% relative 1σ uncertainty on the SH3 Gaussian 3σ flux threshold.

4.4 Mission performance of the final geometry

Figure 10 compares the matched 81-node folds. At 20 d, SG3 accumulates 1635.07 signal and 86806.00 background counts, whereas SH3 accumulates 1678.14 and 15551.68. The resulting Gaussian significances are 5.55 and 13.46; the Asimov values are 5.53 and 13.22.

The SH3 background is 18.10% of the SG3 value while its selected effective area is 99.75% of the reference. The Gaussian 20 d minimum flux is 41.24% of the SG3 threshold, equivalent to a 2.43-fold improvement. The close Gaussian–Asimov agreement at the endpoint confirms that the comparison is not an artefact of the large-background approximation.

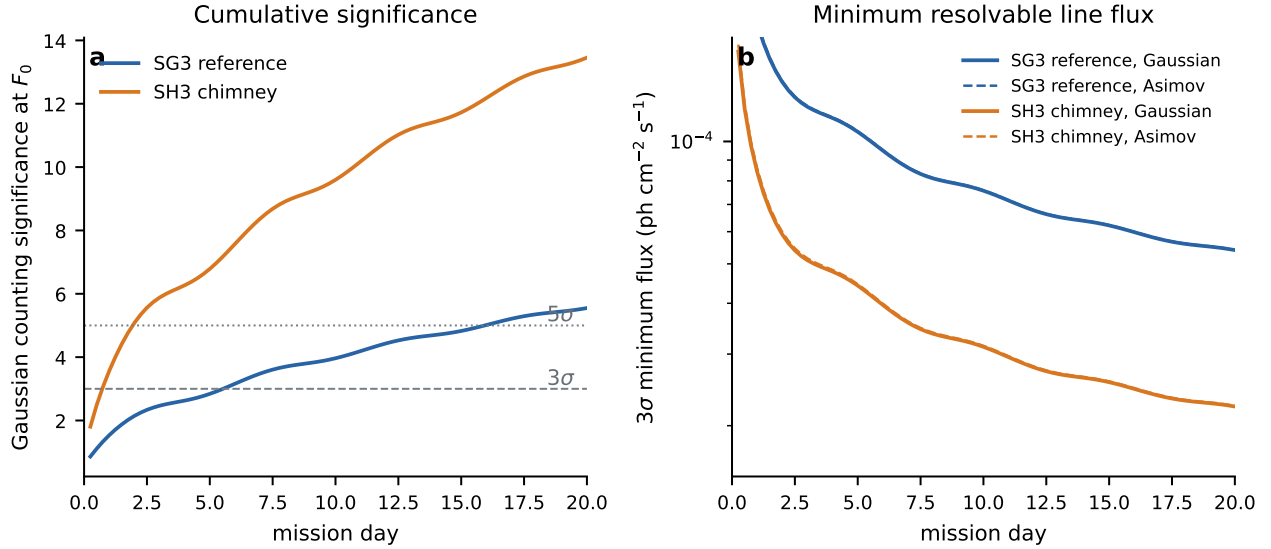


Figure 10: Matched 81-node mission performance for the SG3 reference and SH3 chimney geometries. (a) Gaussian counting significance at $F_0 = 10^{-4} \text{ ph cm}^{-2} \text{s}^{-1}$; horizontal lines mark 3σ and 5σ . (b) Gaussian and Asimov 3σ minimum resolvable flux. Both curves use the same source, response, selection, timing, and atmospheric-transmission contracts.

Table 7: Matched mission projection for the SG3 reference and SH3 chimney geometries. Parentheses give the propagated 1σ statistical uncertainty of the Gaussian 3σ flux threshold.

Quantity	SG3 reference	SH3 chimney
Selected effective area / cm^2	15.12324 ± 0.04492	15.08544 ± 0.04503
Day-15 background / s^{-1}	5.12802×10^{-2}	9.2800×10^{-3}
Day-15 signal at F_0 / s^{-1}	9.53616×10^{-4}	9.79040×10^{-4}
20 d background counts	86806.00	15551.68
20 d signal counts at F_0	1635.07	1678.14
Z_{20d} , Gaussian / Asimov	5.550 / 5.532	13.457 / 13.225
$T_{3\sigma}$, Gaussian / Asimov / d	5.525 / 5.555	0.724 / 0.753
$T_{5\sigma}$, Gaussian / Asimov / d	15.932 / 16.010	1.935 / 2.030
$F_{3\sigma}(20 \text{ d})$, Gaussian / Asimov / $\text{ph cm}^{-2} \text{s}^{-1}$	$(5.406 \pm 0.401) \times 10^{-5} / 5.415 \times 10^{-5}$	$(2.229 \pm 0.209) \times 10^{-5} / 2.238 \times 10^{-5}$

5 Discussion

The main result is a connected physical explanation of the geometry change. In SG3, the TES lies below the staged cold plates, and selected delayed activity from that region supplies 32.18% of the delayed science-window rate. SH3 moves the same six-layer focal plane laterally into a BGO-surrounded chimney, reducing that share to 3.92%. The effective area is unchanged within 0.25%, while the mature background falls by a factor of 5.53. Geometry, selected-event provenance, and mission sensitivity are therefore three views of the same result.

This analysis sequence is consistent with recent instrument-background studies. COSI transports a detailed balloon payload, separates prompt and activation components, and compares their spectral and temporal behaviour with flight data [17]. DIXE likewise begins with a detailed TES payload and asks which incident particles, production regions, and processes make the detected spectrum [18]. Here the distinction between total inventory and selected-event origin is decisive: the leading selected groups are localized copper and near-field structures, so designing from total activity alone would miss the geometric lever.

The final component mix also separates the roles of the instrument layers. Active BGO supplies the dominant suppression, reducing the SH3 direct science-window rate from 2.36263 to $9.8133 \times 10^{-3} \text{ s}^{-1}$. The permissive topology test removes a smaller residual while retaining 98.25% of the science-window signal. Broadband atmospheric gamma supplies the remaining prompt term; activation from all eight incident families supplies the delayed term.

The observing role of this telescope is complementary to INTEGRAL/SPI and COSI, which measure diffuse bulge, disk, and broader Galactic emission [3, 4, 7, 8, 37]. The Laue-lens TES concept exchanges sky coverage

for a concentrated beam and a sub-keV window. Its natural targets are compact 511 keV sources, an unresolved central component, and pointed follow-up observations. The 20 d values in Table 7 quantify that continuously observed, narrow-line reference case.

5.1 Scope and next steps

The mission values are statistical projections for the detector–cryostat geometries and the synthetic, continuously on-source trajectory. They use a 50 keV offline BGO deposited-energy threshold, a detector-level topology test, all eight prompt families, and all eight activation families. The BGO criterion is an ideal deposited-energy threshold rather than a measured hardware trigger efficiency. Background from the Laue optics and balloon platform lies outside the detector–cryostat budget. Radiation-field, cross-section, calibration, reconstruction, and pointing systematics are also outside the propagated statistical uncertainty.

The current precision is limited mainly by the finite number and unequal weights of selected transport events. Future validation should increase the prompt-gamma and dominant delayed-family support, repeat production-position sampling to test spatial convergence, replace the synthetic trajectory with a flight plan and visibility history, and calibrate the energy response and BGO trigger efficiency. A spatial–spectral likelihood can then use the focused spot and control regions while profiling background nuisance parameters [38]. This extends the present source-to-geometry logic without changing the closed SG3–SH3 comparison.

5.2 Conclusions

The SG3 reference identifies the staged cold region and nearby activated structures as a major source of selected delayed background. Relocating the six-layer TES focal plane into the SH3 lateral chimney reduces the cold-stage share from 32.18% to 3.92% while retaining 99.75% of the selected effective area. At day 15, SH3 gives $B = 9.280 \times 10^{-3} \text{ s}^{-1}$ and $S = 9.790 \times 10^{-4} \text{ s}^{-1}$ at $F_0 = 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$. Along the fixed-45° 20 d trajectory it reaches Gaussian and Asimov significances of 13.46 and 13.22, with a Gaussian 3σ threshold of $(2.229 \pm 0.209) \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$. The matched threshold improves by about 2.43 relative to SG3, demonstrating how particle-, nuclide-, and position-resolved background information can be converted into a focal-plane geometry and a mission-scale 511 keV performance prediction.

Data and software availability

Before publication, a versioned public repository or institutional archive should provide the geometry and source definitions, stochastic-analysis conventions, analysis and figure-generation scripts, LaTeX source, and derived validation products needed to reproduce the tables and figures in this paper.

Declarations

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